Federated Intrusion Detection Under Label-Flipping Poisoning: A Simulated Multi-Site Study on CICIoT2023

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I. Problem and Motivation

Retail headquarters, hospitals and bank branches each see attack traffic the others never do, yet privacy policy bars pooling raw captures. Federated learning (FL) lets each site train locally and share only model updates [1]. Since the server never inspects the underlying records, one compromised participant can corrupt the shared model [2]. We quantify that exposure for intrusion detection.

II. Research Question and Expectation

RQ: In a six-client federated IDS trained on non-IID partitions of CICIoT2023, how much global detection performance is lost when one client performs a label-flipping attack, and does robust aggregation recover it?

Threat model and expectation: One client relabels 50% of its attack samples as benign each round; all other parties are honest. Under FedAvg we expect global macro F1 to fall by more than 3 points versus the clean run, and trimmed-mean aggregation [3] to recover at least half of that loss; smaller effects would falsify this.

III. Dataset and Client Partition

We use CICIoT2023 [4], labeled IoT flows from the Canadian Institute for Cybersecurity. Its official page states no license and asks for citation, so we use it for coursework research only (accessed Sep. 20, 2026). It is lab-generated single-testbed data with heavy class imbalance, so the six branch offices are **simulated clients** produced by partitioning, and we make no real-world claims. We build a documented stratified subset of about 300,000 flows, collapsing the 33 attack types into the seven published families plus benign. A stratified 20% test set and 10% validation set are held out first (seed 42); the test set is for final reporting only, and all settings are chosen on validation. The other 70% is grouped into six clients by k-means ($k = 6$, seed 42) on standardized flow features, fit on this pool only, so no labels build the partition; an IID split is the control. Clients are never defined by labels, IPs, ports, timestamps or row indices, and identifier fields (if present) are dropped to prevent shortcut learning. We quantify leakage risk with per-client label distributions and client–label mutual information.

IV. Method and Experimental Conditions

Python, PyTorch and Flower [5] run in single-machine simulation. One small MLP over numeric flow features is held constant; training runs 30 rounds of one local epoch, all six clients participating. Core configurations: (1) a **centralized reference** on pooled data; (2) a **clean federated baseline**, FedAvg on non-IID and IID splits; (3) a **poisoned condition** with one label-flipping client; and (4) **robust aggregation**, trimmed mean, on the poisoned and clean non-IID setups, since honest but heterogeneous clients can look like outliers. Local-only models and median aggregation are extensions.

V. Evaluation Plan

The primary metric is macro F1 on the global test set, since accuracy alone is insufficient on imbalanced data, with per-class precision, recall and FNR, FPR, PR-AUC and confusion matrices. Runs use 3 seeds (mean and standard deviation). Federated runs add per-round performance, client-level dispersion, a communication proxy (bytes per round) and runtime.

VI. Safety, Reproducibility and Timeline

All experiments are simulated on the project's own static data and code; no real service, traffic generation, university or third-party system, or personal data is involved, and poisoning is a controlled label change compared against a clean baseline. The repository will include pinned dependencies, fixed seeds, subset and partition scripts, and a README reproducing every reported number. Milestone 1 (Oct. 25): subset, partitions, centralized model; Milestone 2 (Nov. 15): clean vs. poisoned runs; final (Nov. 30, Dec. 7): limitations, demo, defenses. Members own data preparation, the centralized model, the federated baseline, and attack and defense.

References

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[4] E. C. P. Neto *et al.*, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, 2023. https://www.unb.ca/cic/datasets/iotdataset-2023.html
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